

Task 5: Capture and Analyze Network Traffic Using Wireshark

Objective :

To capture live network traffic using Wireshark and identify different network protocols.

Tools Used :

Wireshark

Procedure :

1. Installed Wireshark.
2. Started packet capture on the active Wi-Fi interface.
3. Browsed websites and generated network traffic.
4. Stopped the capture after one minute.
5. Applied protocol filters (DNS, TCP, TLS).
6. Saved the capture as a .pcap file.

Protocols Identified :

DNS :

Purpose: Resolves domain names into IP addresses.

TCP :

Purpose: Provides reliable communication between devices.

TLS :

Purpose: Encrypts HTTPS communication.

Files Included

network_capture.pcap :

DNS Screenshot

TCP Screenshot

TLS Screenshot

Conclusion :

Successfully captured live network traffic using Wireshark and identified DNS, TCP, and TLS protocols.

